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Method:
For this problem I used the hint to come up with a greedy algorithm that makes sure each index satisfies the condition as it goes. This is a classic greedy approach because it makes the best decision for each pair, and doesn't have to return to any previously checked pairs to ensure correctness of the end result.

Approach:
The program loops through the indices and then for each even index it compares index i and i+1, swapping if i is greater (all even indices should be less than the following index). Similarly for the odd indices it compares i and i+1 and swaps if i+1 is greater (all odd indices should be greater than the following index).

Reasoning:
This algorithm is efficient because it allows for the whole array to be sorted into the pattern in linear time. The loop is O(n) and all of the comparison and swapping is O(1).

Reflection:
I used these test cases:
Example 1:
Input: nums = [3,5,2,1,6,4]
Output: [3,5,1,6,2,4]
Explanation: [1,6,2,5,3,4] is also accepted.

Example 2:
Input: nums = [6,6,5,6,3,8]
Output: [6,6,5,6,3,8]

Time Complexity:
O(n)